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Assessing privacy risks of biometrics

- As the biometric traits are permanent & can't be changed, it should be protected.
- misuse can lead to identity theft and surveillance.

* Types < physiological, behavioural } Each type has diff privacy application

Privacy vs Security

- security protects s/m from attacks.
- privacy protect use identity & personal data
- A secure s/m can still violate privacy.

Key Privacy risks in Biometrics

- Irrevocability of biometric traits.
- Data breaches & unauthorized access
- Function creep & cross matching
- Mass surveillance

Risk: Irrevocability

- Biometric traits can't be replaced like passwords
- compromise causes permanent damage
- Eg: Leaked Fingerprint data

Biometric data breaches

- centralized databases attract attackers.

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- stolen templates can be misused.
- Large scale breaches affect millions.
- Every biometric modality has a unique feature. In fingerprint, the features are patterns. These features are stored as templates in DB.

Function creep

- Data reused beyond original purpose
- often w/o user consent
- Violates data protection principles
- Eg: face recognition details are reused somewhere else

cross matching

- Linking biometric data across s/m
- Enables tracking & profiling
- Reduces anonymity

Surveillance

- facial recognition enables mass surveillance
- used in public spaces w/o consent
- threatens civil liberties.

Assessing privacy risks

- Identify biometric modality used.
- Analyze data collection & storage.

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- Evaluate sharing and access policies
- Assess impact on user rights.
- Every biometric s/ms by their own standards and certain policies are followed by them.

Technical factors affecting privacy
centralized vs decentralized storage
template protection technique
Encryption and access control

Legal and Ethical considerations
user content and transparency
purpose limitation
Compliance with data protection law.

Privacy protection techniques
*) Cancelable biometric, *) Biometric cryptosystem, *) Encrypted & on-device storage.

case study: National Biometric ID s/m
- large scale biometric data collection. High privacy & security risks.
- requires strong safeguards.

Best Practices
- collect minimal biometric data
- use privacy-by-design approach.
- ensure legal compliance & audits.